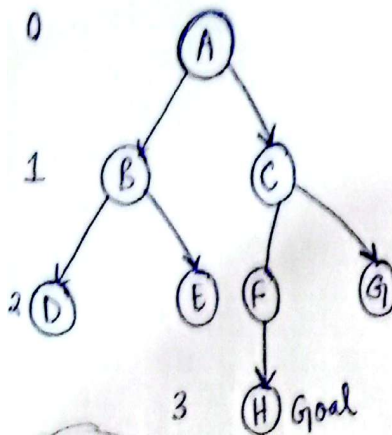


Assignment # 01:

Name: Maabrukh Shakeel
SID: 15975

Q#1: BFS



- ①

A	
---	--
- ②

B	C
---	---
- ③

D	E
---	---
- ④

F	G
---	---
- ⑤

H	
---	--
- ⑥

H	
---	--

- ⑦

H	
---	--
- ⑧

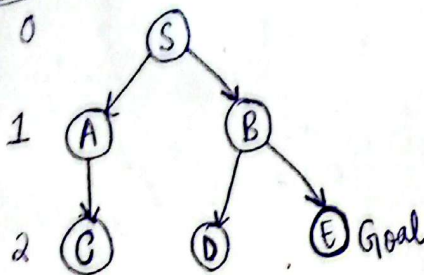
H	
---	--

• Path: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H$

• Completeness: Complete

• Complexity: $O(b^d)$
 $O(2^3) = 8$

Q#2: BFS



- ①

S	
---	--
- ②

A	B
---	---
- ③

C	
---	--
- ④

D	E
---	---

- ⑤

E	
---	--
- ⑥

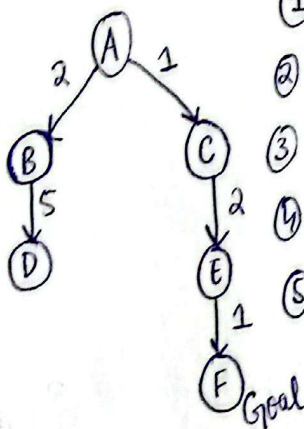
E	
---	--

• Path: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

• Completeness: Complete

• Complexity: $O(b^d)$
 $O(2^2) = 4$

Q#3: UCS



- ①

A(0)

- ②

B(2)	C(1)
------	------
- ③

B(2)	E(3)
------	------
- ④

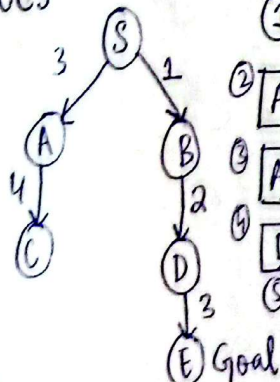
E(3)	D(7)
------	------
- ⑤

D(7)	F(4)
------	------

• Path: $A \rightarrow C \rightarrow B \rightarrow E \rightarrow F$

• Completeness: Complete

Q#4: UCS



- ①

S(0)

- ②

A(3)	B(1)
------	------
- ③

A(3)	D(3)
------	------
- ④

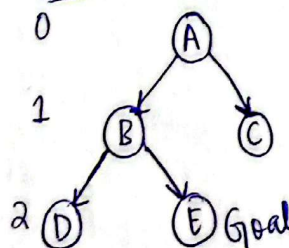
D(3)	C(7)
------	------
- ⑤

C(7)	E(6)
------	------

• Path: $S \rightarrow B \rightarrow A \rightarrow D \rightarrow E$

• Completeness: Complete

Q#5: DFS



- ①

A

- ②

B

- ③

B

- ④

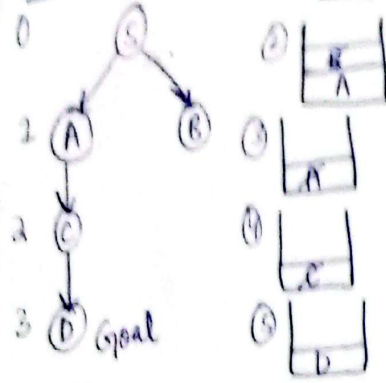
E

• Path: $A \rightarrow C \rightarrow B \rightarrow E$

• Completeness: Complete

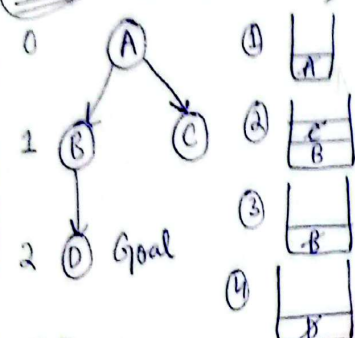
• Complexity: $O(b^d)$
 $O(2^2) = 4$

Q#6: DFS



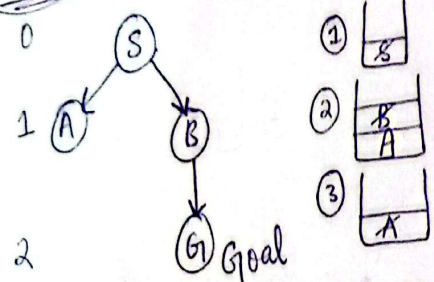
- Path: $S \rightarrow B \rightarrow A \rightarrow C \rightarrow D$
- Completeness: Complete
- Complexity: $O(b^{nd}) \Rightarrow O(2^3) = 8$

Q#7: DLS (limit=2)



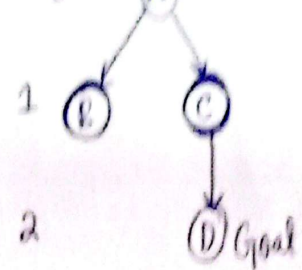
- Path: $A \rightarrow C \rightarrow B \rightarrow D$
- Completeness: Complete
- Complexity: $O(b^{nd}) \Rightarrow O(2^2) = 4$

Q#8: DLS (limit=1)



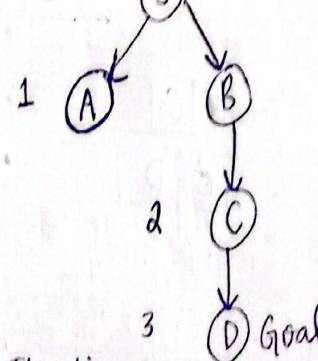
- Path: No path because goal not found.
- Completeness: Not complete
- Complexity: $O(b^{nd}) \Rightarrow O(2^2) = 2$

Q#9: IDS



- Iteration 1: (limit: 0)
- Iteration 2: (limit: 1)
- Iteration 3: (limit: 2)
- Path: $A \rightarrow C \rightarrow D$
- Completeness: Complete
- Complexity: $O(b^{nd}) \Rightarrow O(2^2) = 4$

Q#10: IDS



- Iteration 1: (limit: 0)
- Iteration 2: (limit: 1)
- Iteration 3: (limit: 2)
- Iteration 4: (limit: 3)
- Path: $S \rightarrow B \rightarrow C \rightarrow D$
- Completeness: Complete
- Complexity: $O(b^{nd}) \Rightarrow O(2^3) = 8$